

8_23_2026 Update

Boltz-2, Nesso-1, FlashBind, and an FEP+ four-kinase reproduction

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Executive summary

This week, I focused on four linked questions:

1. What does Boltz-2 do?
2. How does Nesso-1 change that workflow, and what does it improve or sacrifice?
3. Where does FlashBind’s dock-and-score workflow fit relative to these two models?
4. Can I reproduce the reported public affinity benchmark using the authors’ released checkpoints?

In brief, Boltz-2 jointly predicts an atom-resolved protein–ligand complex and learned affinity outputs from protein sequence, ligand chemistry (SMILES), and usually an MSA.

Nesso-1 removes the costly atom-coordinate diffusion stage and the MSA requirement. It instead predicts affinity from jointly processed protein–ligand features and a predicted pocket. This makes Nesso much more practical for screening, at the cost of not producing the detailed heavy-atom pose supplied by Boltz-2.

FlashBind makes a third architectural choice. Rather than jointly generating an atom-resolved complex with an expensive diffusion decoder, it first obtains a three-dimensional receptor, uses FABind+ to predict a pocket and dock the ligand, and then scores the resulting local protein–ligand geometry with an E(3)-equivariant graph neural network. It is therefore best understood as a two-stage *dock then score* system for virtual screening. The receptor can be experimental or predicted, but three-dimensional receptor coordinates are required internally [1].

For the empirical test, I reconstructed the public 87-compound, four-kinase benchmark and ran the released models on every compound. Nesso-1 achieved a compound-count-weighted mean within-target Pearson correlation of $r = 0.807$ (compound-bootstrap 95% interval 0.731–0.873), essentially reproducing the preprint value of 0.80 [2]. Boltz-2, using the MSA-1024 protocol described below, achieved $r = 0.627$ (interval 0.484–0.744), close to its paper reference of 0.66 [3]. FlashBind’s two-checkpoint value ensemble achieved $r = 0.533$ (interval 0.382–0.669), reproducing its paper value of 0.53 [1]. I obtained 87/87 affinity outputs from all three models.

Separately, I compared the available predicted structures with 16 exact experimental protein–ligand cocrystals. Using the same protein-only alignment and symmetry-corrected ligand RMSD definition for both methods, the released FlashBind/FABind+ poses had a median RMSD of **0.634 Å** and 16/16 were below 2 Å. The Boltz-2 MSA-1024 poses had a median of **0.898 Å** and 14/16 were below 2 Å.

Headline result: complete 87-compound benchmark

	Weighted Pearson r	95% interval	Paper reference
Nesso-1	0.807	0.731–0.873	0.80
Boltz-2 MSA-1024	0.627	0.484–0.744	0.66
FlashBind, released poses	0.533	0.382–0.669	0.53

87/87 outputs for all three models • Paired difference: +0.180 [+0.058, +0.319]

Largest target-level separation: p38 ($r = 0.869$ Nesso versus $r = 0.364$ Boltz). The Boltz result uses the documented 1,024-row trunk-MSA subsample; the FlashBind result uses released FABind+ poses.

1 Benchmark

I used the older OpenFF Git commit `da7c337` because it contains the full 87-example benchmark used in both model papers: $16 + 16 + 21 + 34 = 87$ neutral protein–ligand examples [4].

Each target defines one separate medicinal-chemistry series: the protein is held fixed while the ligand is changed. The four targets are cyclin-dependent kinase 2 (CDK2), tyrosine kinase 2 (TYK2), c-Jun N-terminal kinase 1 (JNK1), and p38 α mitogen-activated protein kinase. All four are protein kinases, but they are different proteins with different ligand series and different biochemical assays.

Target / assay series	Ligands (N)	Protein construct input	Endpoint	Reference PDB
CDK2 (cyclin-dependent kinase 2)	16	CDK2, 303 aa; cyclin A2, 258 aa	IC50	1H1Q
TYK2 (tyrosine kinase 2)	16	TYK2, 302 aa	Ki	4GIH
JNK1 (c-Jun N-terminal kinase 1)	21	JNK1, 370 aa; JIP1 peptide, 11 aa	IC50	2GMX
p38 α (MAPK14)	34	p38 kinase, 372 aa	IC50	3FLY

Table 1: Frozen benchmark composition. One model example is one target construct paired with one ligand, so N is the number of independently scored ligands in that target’s series.

The columns have the following precise meanings:

Target protein / assay series. The common protein construct against which a chemically related set of ligands was experimentally tested. I calculated each target’s correlation separately rather than mixing compounds from different targets.

Ligands (N). The number of distinct protein–ligand examples in that series. For example, CDK2 has 16 ligands. Nesso therefore received 16 CDK2 inputs: each contained the same CDK2–cyclin A2 sequences but a different ligand SMILES. The four values sum to $16 + 16 + 21 + 34 = 87$ model examples.

Protein construct. The exact amino-acid sequence or sequences given to the model. “303 aa; 258 aa” means two chains, not one 561-residue protein: a 303-residue CDK2 chain and its 258-residue cyclin A2 partner. Similarly, the JNK1 input includes the 11-residue JIP1 peptide found in the curated reference complex.

Experimental endpoint. The laboratory measurement used as the benchmark’s ground-truth label. IC50 is the concentration that produces 50% inhibition under a particular assay protocol. Ki

is an inhibition constant inferred through an enzyme-kinetic model. Smaller IC50 or Ki means stronger apparent inhibition. These endpoints are related but not interchangeable, and neither should automatically be called a binding free energy. Source values in nM or μM were converted to μM and then to $\log_{10}(\text{value}/\mu\text{M})$ before comparison. CDK2, JNK1, and p38 use IC50 labels; TYK2 uses Ki labels.

Reference PDB. The Protein Data Bank accession for the experimentally solved complex used by the curated benchmark to identify the target and its exact construct. For example, 1H1Q identifies the reference CDK2–cyclin A2 complex. In this calculation, the PDB coordinates and bound ligand pose were *not* given to Nesso or Boltz; only the deposited protein sequence and the benchmark provenance were taken from this record. FlashBind is different: its released-pose calculation uses the corresponding fixed protein coordinates, but not the crystallographic ligand pose.

2 Model description: Boltz-2, Nesso-1, and FlashBind

2.1 Boltz-2: cofold and score

At the highest level, Boltz-2 is a biomolecular cofolding model with an additional affinity-prediction pathway. One example can contain protein sequence(s), ligand chemistry, an MSA, and optional templates or constraints. Its structure trunk integrates these inputs, and a stochastic diffusion decoder generates three-dimensional coordinates for the modeled heavy atoms. The released system returns sampled complex structures and confidence quantities; for protein–ligand inputs it can also return a binder probability and continuous affinity estimates [3]. These learned affinity outputs are assay-calibrated model scores. They are not direct alchemical calculations of an equilibrium ΔG_{bind} .

2.2 Nesso-1: predict affinity without coordinate diffusion

Nesso-1 follows a different affinity workflow. It does not run the atom-coordinate diffusion decoder used by Boltz-2, and it supplies protein context through pretrained ESM-2 embeddings instead of an MSA.

The Nesso paper reports more than a tenfold inference speed-up over Boltz-2, and more than twentyfold on some larger targets, while matching or exceeding Boltz-2 on its evaluated affinity benchmarks [2]. Nesso does not return the atom-resolved complex produced by Boltz-2, so the two models serve different structural needs. For this project, I compare their released affinity outputs on identical protein and ligand inputs. Neither affinity head is a molecular simulation or a direct calculation of ΔG_{bind} ; both are learned from experimental potency and affinity labels.

2.3 FlashBind: dock then score

At the user-facing level, FlashBind can begin with a protein sequence and a ligand SMILES. Internally, however, the sequence must first be resolved to a three-dimensional receptor. The released preprocessing searches for an experimental or predicted structure and can fall back to a structure predictor when one is unavailable. FlashBind then applies the following procedure [1]:

1. **Resolve the receptor.** Use a PDB structure, a predicted structure such as an AlphaFold model, or another generated receptor model. A crystal structure is useful but not mandatory.
2. **Predict the pocket and pose.** FABind+ rapidly identifies a candidate binding region and places the ligand in that region.

3. **Construct the local complex.** The pipeline crops the receptor around the proposed ligand pose and represents the local protein–ligand environment as a geometric graph.
4. **Score the geometry.** An E(3)-equivariant graph neural network processes the local complex. Equivariance ensures that rotating or translating the coordinates does not arbitrarily change the binding prediction.
5. **Read out the prediction.** Separate heads can return a binary activity probability or a continuous assay-derived affinity score. The continuous score is not a thermodynamic calculation of ΔG_{bind} .

This modularity separates pose generation from affinity scoring, which makes large-library screening more practical but also creates a sequential error path:

receptor conformation $\longrightarrow$ pocket selection $\longrightarrow$ docked pose $\longrightarrow$ binding score.

An incorrect receptor or docked pose can therefore limit the scorer even if the final neural network is otherwise well calibrated.

Model workflows at a glance

Boltz-2: sequence + ligand + MSA $\rightarrow$ joint trunk + atom diffusion $\rightarrow$ atom-resolved complex + affinity

Nesso-1: sequence + ligand $\rightarrow$ ESM-2 + pair trunk + predicted pocket $\rightarrow$ affinity, without a final Cartesian complex

FlashBind: 3D receptor + ligand $\rightarrow$ FABind+ pocket and pose $\rightarrow$ local EGNN score

3 Reproducing the benchmark

3.1 Nesso-1

I ran Nesso using the official source at Git revision `f0156e9`, released checkpoint tag `v1.0.0` (resolved Hugging Face revision `1896c84`), and ESM-2 `facebook/esm2_t33_650M_UR50D` at revision `08e4846`. I used seed 42, five recycling steps, bfloat16 mixed precision, the default two-stage 22 Å protein-pocket crop with a 256-token budget, and disabled optional CUDA kernels. I generated each unique protein-chain embedding once per target and reused it across the target’s ligands. I retained the raw continuous affinity values, the two component estimates, binder probabilities, and entropy diagnostics before analysis.

Nesso does not generate an all-atom protein–ligand complex. It combines sequence-derived protein features and ligand atom/bond features in a geometric trunk, predicts an interaction region, and reads out affinity. This is why it can screen faster than a full coordinate-diffusion model [2].

3.2 Boltz-2 inference protocol

I used Boltz 2.2.1 source revision `b1ebfc4`, the released structure and affinity checkpoints, seed 42, three structure recycles, one 200-step structure sample, and five 200-step affinity samples. I generated one ColabFold/MMseqs2 MSA per unique protein chain and reused it across every ligand for that target. I capped each MSA at the frozen maximum of 8,192 parsed rows. I did not use templates, experimental coordinates, or the optional molecular-weight correction.

Normally, all retained MSA rows—up to that 8,192-row cap—can enter the model trunk. That full-depth calculation exceeded the memory of the local 10 GiB GPU, so I enabled Boltz’s documented `--subsample_msa` option and used 1,024 randomly selected MSA rows in the trunk [3]. This reduces memory use while preserving the larger parsed MSA as the sampling pool.

I kept every other setting unchanged and applied this MSA-1024 protocol to all four targets. I obtained 87/87 structures and affinity predictions. This is a released-checkpoint reproduction with reconstructed preprocessing and a documented MSA-depth choice, not the authors’ exact evaluation pipeline.

3.3 FlashBind released-pose scoring run

I used FlashBind source revision `f161268`, the two released value checkpoints at Hugging Face revision `fa6362c`, and the released FEP+4 archive at revision `50b1511`. The archive contains exactly 87 IDs, four fixed receptor PDBs, one FABind+-predicted ligand pose per pair, ESM3 protein representations, and TorchDrug ligand representations. I matched all 87 molecules to my canonical manifest by target and InChIKey. Every experimental label matched to floating-point precision, and the four receptor PDB files were byte-identical to my local benchmark copies.

To remain compatible with the local NVIDIA driver, I used an isolated PyTorch 2.5.1/CUDA 12.1 scoring environment rather than the upstream PyTorch 2.7.1/CUDA 12.6 specification. I retained the authors’ inference code, checkpoint weights, input features, and numerical settings, but this dependency change means the calculation is not a bitwise environment reproduction.

I first ran one CDK2 ligand through both checkpoints and required finite `pred_value`, `pred_value_raw`, and molecular weight fields. After that check passed, I ran all 87 records on one GPU with the authors’ 20 Å local distance threshold and averaged the two checkpoint outputs. I used the calibrated `pred_value`, documented as $\log_{10}(\text{IC}_{50}/\text{Ki}/\text{Kd in } \mu\text{M})$, for the primary analysis while retaining the uncalibrated values. This experiment uses released FABind+ poses rather than rerunning FABind+; it therefore tests the released scorer and supplied preprocessing, not independent end-to-end docking reproducibility.

4 Metric policy

I kept the labels hidden during model execution and joined them only after all native outputs were locked by immutable sample ID. I calculated the primary statistic in two stages:

$$r_t = \text{Pearson}(y_{t,i}, \hat{y}_{t,i}), \quad \bar{r} = \frac{\sum_t n_t r_t}{\sum_t n_t}.$$

Thus, the comparison asks whether each model ranks compounds correctly *within* each kinase series; it does not correlate unrelated assays across targets. Spearman ρ and Kendall τ were secondary ranking metrics.

For error metrics only, I mapped both experimental and predicted $\log_{10}$ micromolar values to an energy-like scale at 298.15 K,

$$G_{\text{eff}} = RT \ln \left(10^{-6} 10^{\log_{10}(C/\mu\text{M})} \right).$$

This is a unit transformation of potency. It must not be interpreted as a thermodynamic standard binding free energy because IC₅₀ depends on assay conditions and Ki is not generally interchangeable with IC₅₀. I report both ordinary MAE and target-centered MAE; centering removes one constant offset per target and better isolates relative ranking/calibration within a series.

I estimated uncertainty with 2,000 bootstrap replicates (seed 20260817). In each replicate, I resampled compounds with replacement separately inside each target and recomputed the compound-weighted

statistic.

4.1 Experimental pose-recovery metric

An experimental structure does not come with an “experimental RMSD.” The experimental ligand coordinates are the reference, and I calculate RMSD between each predicted pose and that reference. I identified 16 exact FEP+4 ligand–target identities with deposited cocrystal structures and used the same reference and metric for FlashBind/FABind+ and Boltz-2.

For each pair, I matched the kinase chains by sequence and defined the experimental pocket as residues having any heavy atom within 5 Å of the crystal ligand. I then fitted predicted to experimental pocket using only matched protein Cα atoms. I held that transform fixed while applying it to the predicted ligand and calculated

$$\text{RMSD}_{\text{pose}} = \min_{\pi \in \mathcal{M}} \sqrt{\frac{1}{N} \sum_{i=1}^N \|R\mathbf{x}_i + \mathbf{t} - \mathbf{x}_{\pi(i)}^{\text{ref}}\|^2},$$

where R and $\mathbf{t}$ come only from the protein fit and $\mathcal{M}$ contains complete bond-order- and chirality-compatible atom mappings. The minimum over $\mathcal{M}$ corrects for chemically equivalent atom permutations. I never fitted the predicted ligand directly to the crystal ligand. I use $\text{RMSD} < 2$ Å only as a conventional descriptive pose threshold, not as a hypothesis-test boundary.

5 Results

Target	Pearson r	Spearman ρ	Kendall τ	Centered MAE (kcal/mol)
CDK2	0.748	0.759	0.550	0.723
TYK2	0.917	0.921	0.767	0.749
JNK1	0.666	0.667	0.482	0.450
p38	0.869	0.894	0.723	0.425
Compound-weighted	0.807	0.819	0.641	0.545

Table 2: Nesso-1 released-checkpoint reproduction. The paper reports a compound-weighted Pearson correlation of 0.80 on this benchmark.

Target	Pearson r	Spearman ρ	Kendall τ	Centered MAE (kcal/mol)
CDK2	0.831	0.865	0.667	0.540
TYK2	0.807	0.779	0.617	0.663
JNK1	0.758	0.805	0.616	0.442
p38	0.364	0.433	0.287	0.781
Compound-weighted	0.627	0.666	0.497	0.633

Table 3: Boltz-2 MSA-1024 released-checkpoint reproduction. The paper references are weighted Pearson $r = 0.66$ and Kendall $\tau = 0.48$; the present values are 0.627 and 0.497.

Target	Pearson r	Spearman ρ	Kendall τ	Centered MAE (kcal/mol)
CDK2	0.612	0.641	0.433	0.754
TYK2	0.134	0.147	0.083	1.021
JNK1	0.546	0.578	0.425	0.509
p38	0.677	0.591	0.413	0.611
Compound-weighted	0.533	0.515	0.359	0.688

Table 4: FlashBind two-checkpoint ensemble using the authors’ released FABind+ poses. The reported references are weighted Pearson $r = 0.53$ and Kendall $\tau = 0.38$; the present values are 0.533 and 0.359.

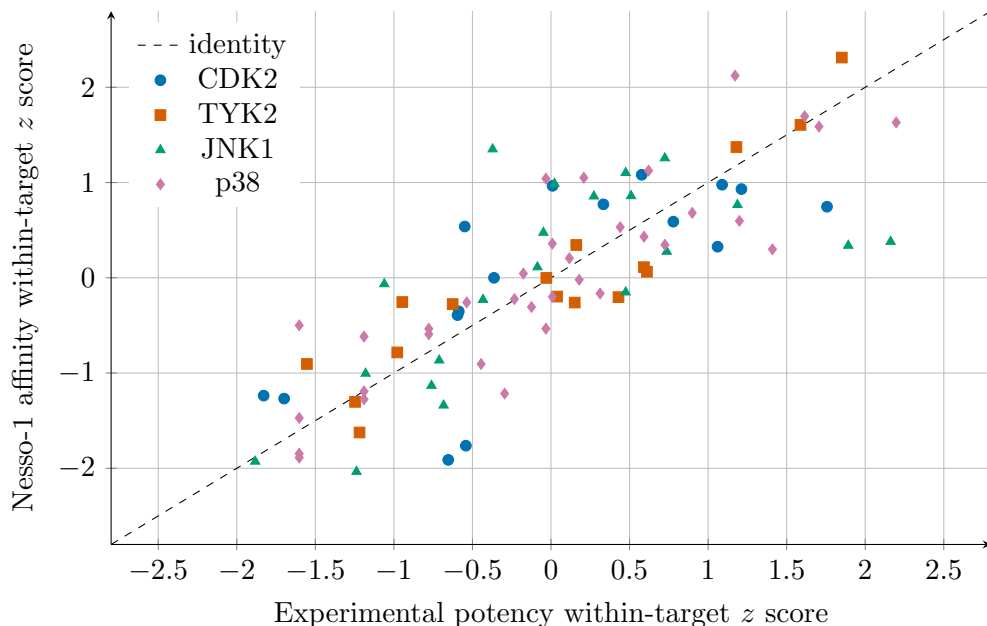


Figure 1: Within-target standardized experimental and predicted potency. Lower values correspond to stronger binding on both axes. Standardization removes target-specific offsets and scale differences for visualization only; metrics were computed from the unstandardized values.

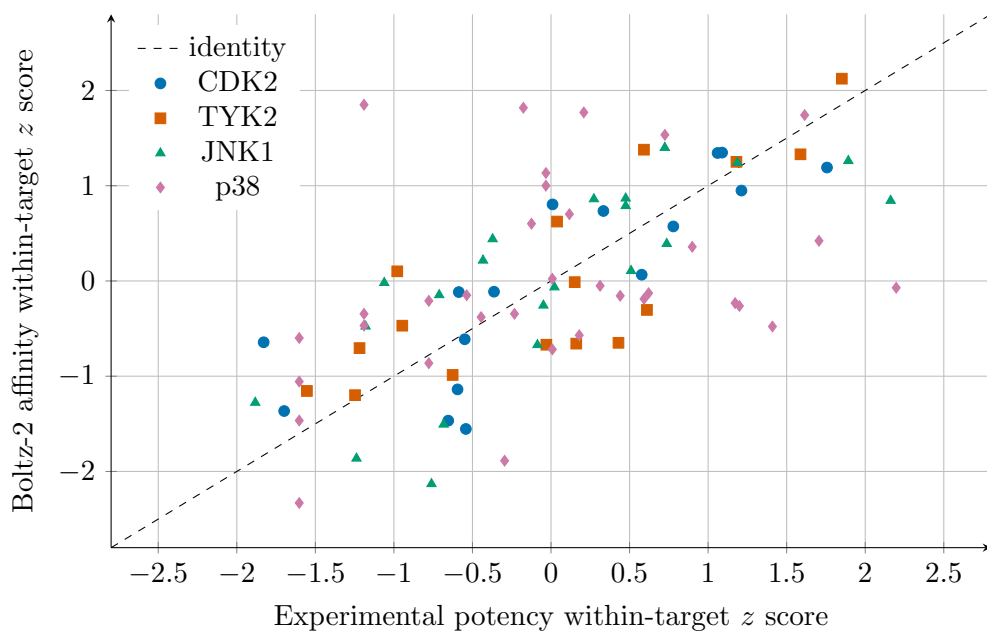


Figure 2: Boltz-2 MSA-1024 standardized scores. The dispersed p38 series is the main target-level departure from Nesso-1 and from the other Boltz targets.

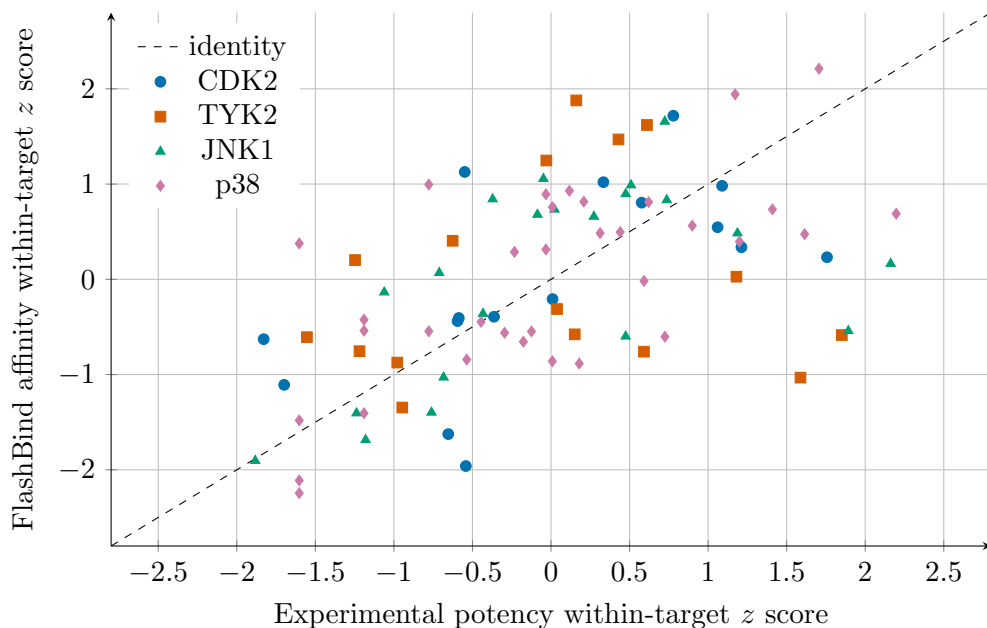


Figure 3: FlashBind standardized scores from the released FABind+ poses. TYK2 is the clear weak target, whereas the other three series retain positive within-target ranking relationships.

I obtained an aggregate Pearson result of 0.807 for Nesso (95% interval 0.731–0.873), which agrees closely with the reported 0.80. For Boltz, I obtained 0.627 (interval 0.484–0.744), which is 0.033 below the paper reference of 0.66; its Kendall result of 0.497 is 0.017 above the paper reference of 0.48. Given reconstructed preprocessing, one random seed, and an explicit MSA-depth change, I regard this as a close released-model reproduction rather than numerical identity.

For FlashBind I obtained weighted Pearson $r = 0.533$ (95% interval 0.382–0.669), essentially identical

to the reported 0.53. The weighted Kendall value was 0.359, 0.021 below the reported 0.38. The close Pearson match is reassuring for software reproduction, but it is not independent validation: the authors supplied the FABind+ poses and representations used by their own scoring benchmark.

Statistic	Nesso-1	Boltz-2	
		MSA-1024	FlashBind
Pearson r	0.807	0.627	0.533
Spearman ρ	0.819	0.666	0.515
Kendall τ	0.641	0.497	0.359
Centered MAE (kcal/mol)	0.545	0.633	0.688

Table 5: Full matched comparison across the 87-compound benchmark. Each entry is the compound-count-weighted mean of the corresponding within-target statistic.

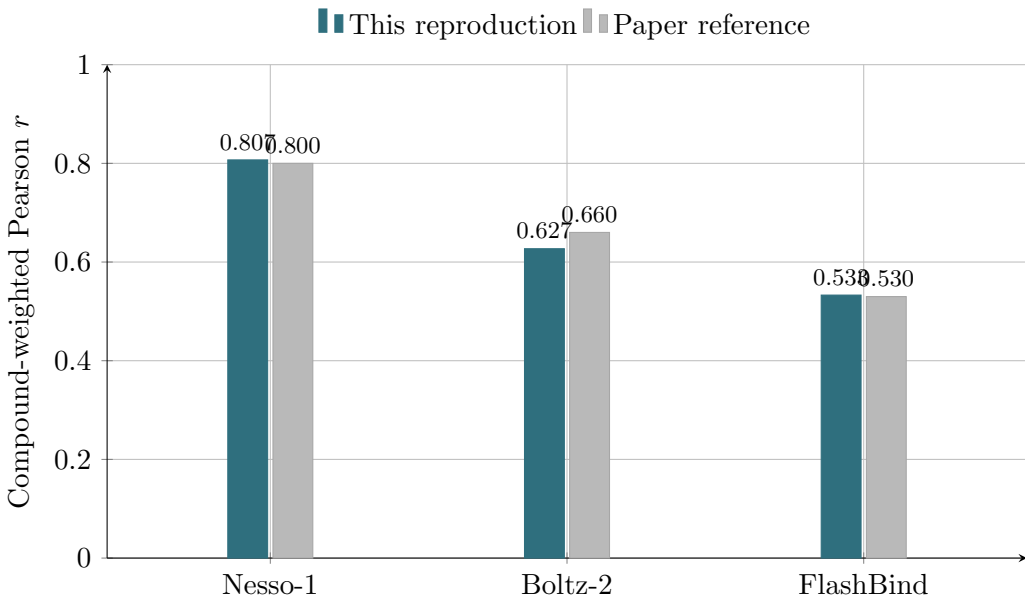


Figure 4: Weighted within-target Pearson correlations from this reproduction and the corresponding paper reference values. The reproduced values differ from the references by +0.007 for Nesso-1, -0.033 for Boltz-2, and +0.003 for FlashBind. Bars show point estimates; the bootstrap intervals are reported in the text and executive-summary table.

I found that the paired interval excludes zero for all three correlation statistics. It barely includes zero for centered MAE, so the evidence is stronger for a ranking difference than for an offset-corrected error difference. Target-level heterogeneity is scientifically important: Boltz is better on CDK2 and JNK1, Nesso is better on TYK2, and p38 supplies the largest Nesso advantage ($r = 0.869$ versus 0.364). My experiment does not determine whether that p38 difference arises from model architecture, MSA subsampling, checkpoint stochasticity, or reconstructed input/preprocessing details. FlashBind shows a different pattern: p38 is its strongest target ($r = 0.677$), while TYK2 is much weaker ($r = 0.134$). Because this run uses released docked poses, that TYK2 result cannot be diagnosed as a fresh docking failure without separately examining the archived poses.

5.1 Experimental pose recovery on the 16 structure-matched pairs

Pose source	Receptor input	Median RMSD (Å)	Mean RMSD (Å)	< 2 Å
FlashBind/FABind+, released poses	Fixed experimental 3D	0.634	0.651	16/16
Boltz-2 MSA-1024, seed 42	Sequence + MSA	0.898	0.977	14/16

Table 6: Symmetry-corrected ligand heavy-atom RMSD after aligning only protein pocket $C\alpha$ atoms. Both rows use the same 16 exact cocrystal references and scoring code, but the receptor inputs are materially different.

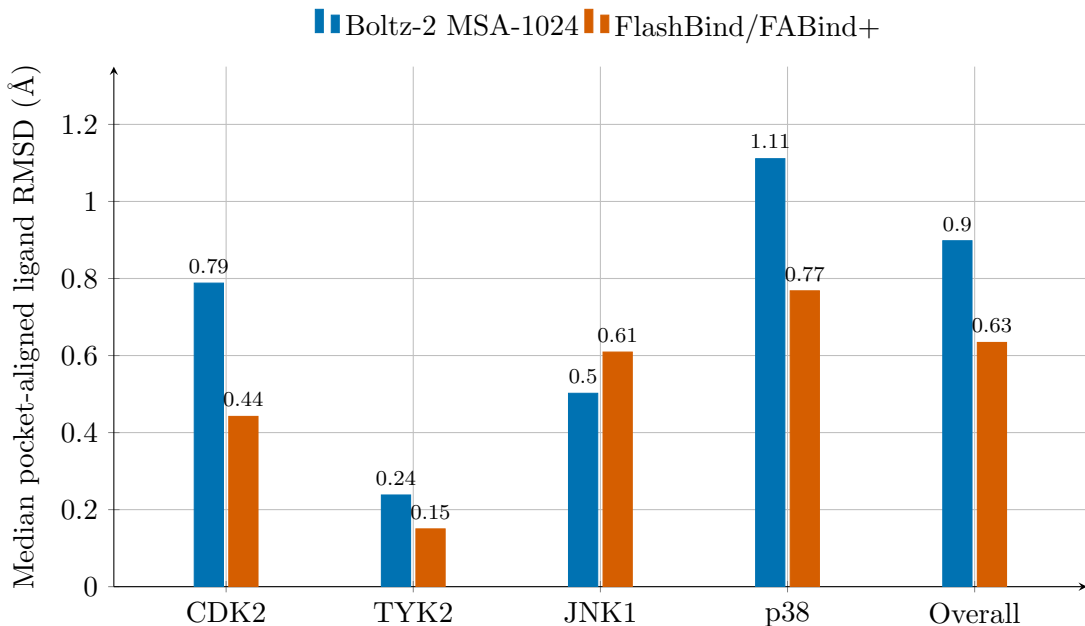


Figure 5: Median symmetry-corrected ligand RMSD by target and over all 16 structure-matched pairs. The target counts are CDK2 $n = 6$, TYK2 $n = 2$, JNK1 $n = 1$, and p38 $n = 7$; the JNK1 bar is therefore a single pair rather than a stable target-level estimate. FlashBind has the lower overall median, but the per-pair comparison remains mixed (9 lower for FlashBind and 7 for Boltz-2).

The released FlashBind/FABind+ poses were below 2 Å for all 16 pairs, whereas the Boltz-2 poses were below 2 Å for 14. FlashBind had the lower RMSD on 9 pairs and Boltz-2 on 7. The median paired FlashBind-minus-Boltz difference was -0.109 Å; its mean was -0.326 Å, influenced by the two Boltz poses above 2 Å. This pattern does not support a claim of uniform FlashBind superiority.

I also audited the coordinate identities before scoring. Every released SDF matched its reference ligand in heavy-atom count and admitted a complete chemical graph mapping. The released p38 receptor has a formatting defect in which the chain identifier disappears after residue 168; I joined the two continuously numbered fragments in memory without changing coordinates and recorded the repair. The archive documents these ligands as FABind+ docking poses, but it does not contain the original docking commands, seeds, or per-pose logs. Thus I have evaluated the supplied poses, not independently reproduced their generation.

6 Interpretation and next-week plan

Using independently reconstructed inputs, I reproduced the Nesso-1 and Boltz-2 paper headlines to within 0.007 and 0.033 Pearson units, respectively. Using the authors’ released pose archive, I reproduced FlashBind to within 0.003 Pearson units. I view these as useful but differently strong reproducibility results: FlashBind’s input preprocessing is less independent. My paired comparison also supports higher average within-series ranking correlations for Nesso on these exact 87 examples, but it does not establish universal superiority or show that the model learned transferable physics. The benchmark is small, public, and congeneric; paper-level train/test chemical overlap may matter, and the Boltz run uses a documented 1,024-row trunk-MSA subsample.

The 16-pair structure analysis adds a complementary result: both methods often recover the known pose on this selected public subset, and the released FlashBind/FABind+ archive has the lower aggregate RMSD. However, FlashBind was given fixed experimental receptor structures while Boltz-2 had to generate the receptor–ligand complex. I therefore interpret the pose result as a successful archive and metric validation, not a fair end-to-end ranking of the two architectures.

Next week, I will move from benchmark reproduction to a source-code-level examination of how Boltz-2, Nesso-1, and FlashBind represent protein–ligand interactions. I will use the TYK2–4GIH complex as a common worked example and trace one forward pass through each released model, recording the residue, atom, and pair representations that connect the inputs to the affinity output.

I will then perform one small probe matched to each architecture:

- for Boltz-2, compare fixed-seed predictions as MSA information is removed or subsampled;
- for Nesso-1, compare the response to sequence changes near its predicted pocket with changes far from that pocket; and
- for FlashBind, compare experimental, released FABind+, displaced, and clashing ligand poses while also checking whole-complex rigid-body invariance.

The purpose is not to establish another performance ranking. It is to determine what information each affinity head actually receives and whether its behavior is consistent with the invariances, locality, and geometric sensitivity expected for protein–ligand binding. End-to-end FABind+ docking, additional stochastic replicates, and the p38 target-level discrepancy remain useful later experiments once this architecture audit is in place.

Reproducibility record

The tracked manifest SHA-256 is

5f0fb23bb97ea84f3da153e050d5913b97da9bba091fb84cd556b9894fdac1e1.

The Nesso checkpoint SHA-256 is

9928a8a824d147d665e76656804af1cd91c86731516d0b561f9fd1c91ee45622.

The Boltz structure and affinity checkpoint SHA-256 values are

Structure: 090e82ac8c92f5e943fa1b39e7410a44027bea7243c0bbb3caa67a77fc1428e1
Affinity: dcc5cd3722b1c9eaa34267e4ae32f55cbbf1963f4c19319381ccfa30fdd2ca9e.

The two FlashBind value-checkpoint SHA-256 values are

Value 1: ca87f84dedd4642da693a508b1119c802caa370e356dad31c47b8a708821e553
Value 2: 59fa2686b0859a632862aec21ea69f2bebb1ab104e48606fe739c929368d488d.

All commands, environments, timestamps, hardware samples, return codes, raw outputs, and parsing status are stored beneath the ignored experiment run tree. Compact metrics and this report are tracked in Git.

References

- [1] Songlin Jiang, Yifan Chen, Aarti Krishnan, Yu Zhang, and Wengong Jin. Flashbind: Towards accurate and efficient structure-based virtual screening. *bioRxiv*, 2026. doi: 10.64898/2025.12.22.695983. URL <https://doi.org/10.64898/2025.12.22.695983>. Preprint, version 2, posted 8 April 2026; version 1 was titled FlashAffinity.
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- [4] David F. Hahn et al. Best practices for constructing, preparing, and evaluating protein-ligand binding affinity benchmarks. *Living Journal of Computational Molecular Science*, 4(1):1497, 2022. doi: 10.33011/livecoms.4.1.1497. URL <https://doi.org/10.33011/livecoms.4.1.1497>.